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RESEARCH INTERESTS

Interested in building trustworthy Machine Learning systems and developing methods for verifying its properties. Passionate of formalizing general problems as optimization problems and finding ways of solving them. Motivated by the intersection of theory and implementation and its fast adaptation into society.

EDUCATION

- EPFL** **PhD in Robustness of Machine Learning** **09/23 - Today**
LIONS (Volkan Cevher's Lab) - École Polytechnique Fédérale de Lausanne (EPFL)
- uc3m** **MSc in Applied and Computational Mathematics** **09/22 - 06/23**
Universidad Carlos III de Madrid (UC3M)
-  **BSc in Data Science and Engineering** **09/18 - 07/22**
Universitat Politècnica de Catalunya (UPC)

WORK EXPERIENCE

-  **Student Researcher** **5/25 - Today**
Google Deepmind Zürich
- Working with the Universal Knowledge Base team on Retrieval Augmented Generation (RAG) research. Studying methods to organize knowledge and make it accessible to Large Language Models (LLMs).
-  **Research Intern** **6/21 - 1/22**
Institut de Robòtica e Informàtica Industrial (IRII) Barcelona
- Worked with Dr. Jordi Sánchez Riera from on building a system able to classify different helicopter models, read their plates and track their movement and the people in the heliport in real time. Successfully delivered the product to the company requesting it.

RESEARCH EXPERIENCE - PUBLICATIONS

- Robustness in Both Domains: CLIP Needs a Robust Text Encoder**   **10/24 - 6/25**
PhD Project / Under submission [ArXiv](#)
- Certified Robustness Under Bounded Levenshtein Distance**  **2/24 - 1/25**
PhD Project / ICLR'25 [ArXiv](#)
- Membership Inference Attacks against Large Vision-Language Models**  **1/24 - 8/24**
PhD Project / NeurIPS'24 [ArXiv](#)
- Revisiting Character-level Adversarial Attacks for Language Models**  **9/23 - 2/24**
PhD Project / ICML'24 [ArXiv](#)
- Efficient local linearity regularization to overcome Catastrophic Overfitting**  **2/23 - 9/23**
Master's Thesis / ICLR'24 [ArXiv](#)

AWARDS

- **Winner of the [ETHz Datathon](#)** (2025)
- **Winner of the BitsXlaMarató Hackathon** (2020)
- **Winner of the FME Datathon** (2021)
- **First place** in the Elche Regional Physics Olympiad (2017)
- **Second place** in the Elche Regional Mathematics Olympiad (2017)
- **Participant** in the Spanish National Physics Olympiad (2018)
- **JAE-Intro Introduction to Research Scholarship.** Awarded by Spanish National Research Institute (CSIC) to excellent BSc and MSc students (2021)
- **Honors in the Statistical Signal Processing course** (2023)
- **Honors in the MSc Thesis** (2023)

SKILLS

Programming	Python, R, C, C++, Git, bash, $\text{\LaTeX}$, Matlab
Communication	Spanish (native), Catalan (advanced), English (advanced), French (beginner)

HOBBIES

Outdoors and endurance sports enthusiast. Competitor in Track and Field and Trail Running at top national level for the last 7 years.

Shortlist of Accomplishments

- **Mont Blanc (4810m) ascensionist** (Chamonix, 2015).
- **10th in the Spanish Vertical Kilometer Championships** (Fuente Dé, 2016)
- **6th in the Spanish Trail Running Championships** (Zumaia, 2017)
- **Spanish High Performance Athlete Level B** (2017-2019)
- **2nd in the Valencian Vertical Kilometer Championships** (Finestrat, 2021)
- **18th team in the Spanish Cross Country Championships by Teams** (Atapuerca, 2022)